Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The mean deviation of an ungrouped data is 50 . If each observation is increased by 2%, then the new mean deviation is

(A) 50

(B) 51

(C) 49

(D) 50.5

Q.2

The mean of variable $1, 2, \ldots n$ whose corresponding frequencies are $1, 2, \ldots n$ is given by

(A) $\frac{n+1}{2}$

(B) $\frac{2n+1}{6}$

(C) $\frac{n(n+1)}{2}$

(D) $\frac{2n+1}{3}$

Q.3

The mean of 5 observations is 4.4 and the variance is 8.24. If three of the five observations are 1,2 and 6, the two values are

(A) 4 and 9

(B) 3 and 5

(C) 2 and 6

(D) 4 and 6

Q.4

In a set of 20 observations, each of the observation below the median of all observations is increased by 6 and each of the remaining observation is decreased by 2. Then the mean of the new set of observations

(A) increase by 1

(B) decrease by 1

(C) increase by 4

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(D) increase by 2

Q.5

Let G_1, G_2 be the geometric means of two series $x_1, x_2, \ldots x_n; y_1, y_2 \ldots y_n$. If G is the geometric mean of $x_i/y_i, i = 1, 2, \ldots n$, then G is equal to

(A) $G_1 - G_2$

(B) $\log G_1 / \log G_2$

(C) $\log (G_1/G_2)$

(D) none of these

Q.6

If $\sum_{i=1}^{18} (x_i - 8) = 9$ and $\sum_{i=1}^{18} (x_i - 8)^2 = 45$ then the standard deviation of $x_1, x_2 \ldots x_{18}$ is

(A) $4/9$

(B) $9/4$

(C) $3/2$

(D) none of these

Q.7

If a variable takes the values $0, 1, 2 \ldots n$ with frequencies proportional to the binomial coefficients ${}^nC_0, {}^nC_1, \ldots, {}^nC_n$ then the mean of the distribution is

(A) $\frac{n(n+1)}{4}$

(B) $\frac{n}{2}$

(C) $\frac{n(n-1)}{2}$

(D) $\frac{n(n+1)}{2}$

Q.8

The mean of two samples of sizes 200 and 300 were found to be 25,10 respectively. Their standard deviations were 3 and 4 respectively. The variance of combined sample of size 500 is

(A) 64

(B) 65.2

(C) 67.2

(D) 64.2

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Q.9

If the mean deviation of the numbers 1, $1+d, \ldots, 1+100d$ from their mean is 255, then d is equal to

(A) 10.1

(B) 20.2

(C) 10.0

(D) 20.0

Q.10

Consider the following statements:
(1) Mode can be computed from histogram
(2) Median is not independent of change of scale
(3) Variance is independent of change of origin and scale
Which of these is/are correct?

(A) Only (1) and (3)

(B) Only (2)

(C) Both (1) and (2)

(D) $(1), (2), (3)$

Q.11

If in a frequency distribution, the mean and median are 21 and 22 respectively, then the mode is approximately

(A) 25.5

(B) 24.0

(C) 22.0

(D) 20.5

Q.12

A scientist is weighing each of 30 fishes. Their mean weight worked out is 30gm and a standard deviation of 2gm. Later, it was found that the measuring scale was misaligned and always under reported every fish weight by 2gm. The correct mean and standard deviation (in gm) of fishes are respectively:

(A) 32,2

(B) 32,4

(C) 28,2

(D) 28,4

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Q.13

The mean and the median of 100 observations have been computed to be 60 and 70 respectively. Later it was discovered that three observations which have been recorded 18,28 and 98 are actually 80,26 and 38 respectively. If the mean and median are recalculated with actual observations, then

(A) median will change but mean will not change

(B) neither mean nor median will change

(C) both mean and median will change

(D) mean will change but median will not change

Q.14

If A.M., G.M. and H.M. in any series are equal then

(A) the distribution is symmetric

(B) all the values are same

(C) the distribution is unimodal

(D) none of these

Q.15

If $n = 10, \bar{x} = 12$ and $\Sigma x^2 = 1530$, then coefficient of variation is

(A) 16

(B) 9

(C) 25

(D) 36

Q.16

The mean square deviation of a set of n observations $x_1, x_2, \ldots, x_n$ about a point c is defined as $\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^2$ The mean square deviation about -2 and 2 are 18 and 10 respectively, then standard deviation of this set of observation is

(A) 3

(B) 2

(C) 1

(D) none of these

Q.17

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$1, 2, 4, 6, 8, 9$ transformed into another series whose mean is 12. Two middle terms of the transformed series are

(A) 7,8

(B) 10,14

(C) 9,15

(D) 11,13

Q.18

The geometric mean of $3, 3^2 \ldots, 3^n$ is

(A) $3^{n/2}$

(B) $3^{(n+1)/2}$

(C) $3^{n(n+1)/2}$

(D) none of these

Q.19

Given the following frequency distribution with some missing frequencies

Class	Frequency
$10-20$	180
$20-30$	$--$
$30-40$	34
$40-50$	180
$50-60$	136
$60-70$	$--$
$70-80$	50

If the total frequency is 685 and median is 42.6 then the missing frequencies are

(A) 81,24

(B) 80,25

(C) 82,23

(D) 832,22

Q.20

In any discrete series (when all the values are not same) the relationship between M.D. about mean and S.D. is

(A) M.D. = S.D.

(B) M.D.> S.D.

(C) M.D. < S.D.

(D) M.D. $\leq$ S.D.

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Q.21

If the standard deviation of $x_1, x_2, \ldots, x_n$ is 3.5, then the standard deviation of $-2x, -3, -2x_2, -3 \ldots, -2x_n - 3$ is

Q.22

If the median of 21 observations is 40 and if the observations greater than the median are increased by 6 then the median of the new data will be

Q.23

A factory is operating in two shifts, day and night, with 70 and 30 workers respectively. If per day mean wage of the day shift workers is 54 and per day mean wage of all the workers is Rs. 60, then per day mean wage of the night shift (in Rs.) is:

Q.24

In a class of 20 students, each student can score either 10 or 0 marks in a certain examination. The maximum possible variance in the marks of the students in the class is

Q.25

If the mean and the standard deviations of 10 observations $x_1, x_2 \ldots x_n$ are 2 and 3 respectively, then the mean of $(x_1+1)^2, (x_2+1)^2 \ldots (x_{10}+1)^2$ is equal to

Answers & Solutions

Q.1 Answer:

51

Solution:

If $x_1 \ldots x_n$ are the observations then the new observations are (1.02) $x_1, (1.02)x_2.., (1.02)x_n$

Therefore the new mean $= (1.02)\bar{x}$

$$\begin{aligned}\text{New mean deviation} &= \frac{1}{n}\sum |(1.02)x_i - (1.02)\bar{x}| \\ &= (1.02)\frac{1}{n}\sum |x_i - \bar{x}| \\ &= (1.02) \times 50 = 51\end{aligned}$$

Q.2 Answer:

$\frac{2n+1}{3}$

Solution:

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Mean $= \frac{1.1+2.2+\ldots n \cdot n}{1+2\ldots+n}$

$= \frac{\sum_{K=1}^{n} k^2}{\sum_{K=1}^{n} k} = \frac{n(n+1)(2n+1)}{6} \times \frac{2}{n(n+1)}$

$= \frac{2n+1}{3}$

Q.3 Answer:

4 and 9

Solution:

$\Sigma x_i = n\bar{x} = 5 \times 4.4 = 22$

So $1 + 2 + 6 + x_4 + x_5 = 22$

$x_4 + x_5 = 13$ **(1)**

$$\sigma^2 = \frac{1}{n}\sum x_1^2 - \left(\frac{1}{n}\sum x_i\right)^2 \Rightarrow \sum x_i^2 = n\left(\sigma^2 + \bar{x}^2\right)$$

$$= 5(8.24 + 19.36)$$

$$= 138$$

$\Rightarrow 1 + 4 + 36 + x_4^2 + x_5^2 = 138$

$\Rightarrow x_4^2 + x_5^2 = 97$

So $x_4^2 + (13 - x_4)^2 = 97 \Rightarrow x_4^2 + 169 + x_4^2 - 26x_4 = 97$

$\Rightarrow 2x_4^2 - 26x_4 + 72 = 0$

$\Rightarrow x_4^2 - 13x_4 + 36 = 0$

$x_4 = \frac{13 \pm \sqrt{169-144}}{2} = \frac{13 \pm 5}{2} = 9, 4$

So $x_5 = 13 - x_4 = 4, 9$

Hence, the other two nembers are 4 and 9.

Q.4 Answer:

increase by 2

Solution:

Arrange all observations in increasing or decreasing order.

Let the observations be $x_1, x_2 \ldots x_{20}$ **so that** $\frac{x_{10}+x_1}{2}$ **is the median. The new set of observations will be** $x_1 + 6, \ldots x_{10} + 6, x_{11} - 2, \ldots, x_{20} - 2$ **Their mean is equal to**

$$\frac{1}{20}\left(\sum_{i=1}^{20} x_i + 6 \times 10 - 2 \times 10\right)$$

$$= \frac{1}{20}\sum_{i=1}^{20} x_i + 2$$

Q.5 Answer:

none of these

Solution:

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Given series $x_1, x_2, \ldots x_n$ -

Geometric mean $G_1 = (x_1 \times x_2 \times \ldots x_n)^{1/n}$ **....(1)**

Another series $y_1, y_2 \ldots, y_{n*}$

Geometric mean $G_2 = (y_1 \times y_2 \times \ldots y_n)^{1/n}$ **....(2)**

Also given series $= \frac{x_1}{y_1}\frac{x_2}{y_2} \ldots \frac{x_a}{y_n}$

Geometric mean $G = \left(\frac{x_1}{y_1} \times \frac{x_2}{y_2} \times \cdots \times \frac{x_n}{y_n}\right)^{1/n}$

$\Rightarrow G = \frac{(x_1 \times x_2 \times x_3 \cdots \times x_n)^{1/a}}{(y_1 \times y_2 \times y_3 \cdots \times y_n)^{1/a}}$

$\Rightarrow G = \frac{G_1}{G_2} \quad (\text{by}(1)\text{and}(2))$

Q.6 Answer:

$3/2$

Solution:

Let $d_i = x_i - 8$ **but**

$$\sigma_x^2 = \sigma_d^2 = \frac{1}{18}\sum d_i^2 - \left(\frac{1}{18}\sum d_i\right)^2$$

$$= \frac{1}{18} \times 45 - \left(\frac{9}{18}\right)^2 = \frac{5}{2} - \frac{1}{4} = \frac{9}{4}$$

Therefore $\sigma_x = 3/2$

Q.7 Answer:

$\frac{n}{2}$

Solution:

$N = \Sigma f_i = k\left[{}^nC_0 + {}^nC_1 + \ldots + {}^nC_n\right] = k(1+1)^n$

$= k2^n$

$\Sigma f_i x_i = k\left[1.{}^nC_1 + 2 \cdot {}^nC_2 + \ldots + n{}^nC_n\right]$

$= kn\left[1 + (n-1) + \frac{(n-1)(n-2)}{2!} + \ldots + 1\right] = kn(1+1)^{n-1}$

$= kn2^{n-1}$

Thus $\bar{x} = \frac{1}{2^n}\left(n2^{n-1}\right) = \frac{n}{2}.$

Q.8 Answer:

67.2

Solution:

Combined mean $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$= \frac{200 \times 25 + 300 \times 10}{500} = 16$

Let $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$ and

$d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

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Now we know that

$$\sigma^2 = \frac{n_1\left(\sigma_1^2+d_1^2\right)+n_2\left(\sigma_2^2+d_2^2\right)}{n_1+n_2}$$
$$= \frac{200(9+81)+300(16+36)}{500} = \frac{33600}{500} = 67.2$$

Q.9 Answer:

10.1

Solution:

$$\bar{x} = \frac{1}{101}[1+(1+d)+(1+2d)+\ldots..+(1+100d)]$$
$$= \frac{1}{101} \times \frac{101}{2}[1+(1+100d)] = 1+50d$$

$\therefore$ **Mean deviation from mean**

$$= \frac{1}{101}[11-(1+50d)|+|1+d-$$
$$(1+50d)|+\ldots..+|1+100d-(1+50d)\ |]$$
$$= \frac{2|d|}{101}\{1+2+\ldots.+50]$$

Q.10 Answer:

Both (1) and (2)

Solution:

It is true that mode can be computed from histogram and median is not independent of change of scale

But variance is independent of change of origin and not of scale.

Q.11 Answer:

24.0

Solution:

Mode $= 3$ **Median** -2 **mean**

$= 3 \times 22 - 2 \times 21$

$= 24$

Q.12 Answer:

32,2

Solution:

Given that mean weight of fishes is 30gm

Standard deviation is 2gm

Given that every fish weight is increased by 2gm

Therefore, over all mean weight is $30+2 = 32\text{gm}$

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Standard deviation is not effected by the addition of a constant. Therefore standard deviation remains same.

Q.13 Answer:

median will change but mean will not change

Solution:

Corrected mean.

$= \frac{1}{100}[60 \times 100 - 18 - 28 - 98 + 80 + 26 + 38]$

$= 60$ **(same as the original mean)**

Median will change (see theory).

Q.14 Answer:

all the values are same

Solution:

A.M = G.M ... Given

So, $\frac{a+b}{2} = \sqrt{ab} \Rightarrow (a+b)^2 = 4ab \quad \ldots$

Squaring on both sides $\Rightarrow a^2 + 2ab + b^2 = 4ab$

$\Rightarrow a^2 - 2ab + b^2 = 0$

$\Rightarrow (a-b)^2 = 0 \Rightarrow a = b$

$\text{A.M} = \text{H.M.} \quad \text{...Given}$

So, $\frac{a+b}{2} = \frac{2ab}{a+b}$

$\Rightarrow (a+b)^2 = 4ab$

$\Rightarrow a^2 + 2ab + b^2 = 4ab$

$\Rightarrow a^2 - 2ab + b^2 = 0$

$\Rightarrow (a-b)^2 = 0$

$\Rightarrow a = b$

$\text{G.M} = \text{H.M.} \quad \text{....Given}$

So, $\sqrt{ab} = \frac{2ab}{a+b}$

$\Rightarrow a + b = 2\sqrt{ab}$

$\Rightarrow (a+b)^2 = 4ab \quad$ squaring on both sides

$\Rightarrow a^2 + 2ab + b^2 = 4ab$

$\Rightarrow a^2 - 2ab + b^2 = 0$

$\Rightarrow (a-b)^2 = 0$

$\Rightarrow a = b$

Q.15 Answer:

25

Solution:

$\sigma^2 = \frac{1}{n}\Sigma x^2 - \bar{x}^2 = \frac{1530}{10} - 12^2 = 9$

$\text{C.V} = 100 \times \frac{\sigma}{\bar{x}} = 100 \times \frac{3}{12} = 25$

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Q.16 Answer:

3

Solution:

Given that $18=\frac{1}{n}\sum\left(x_i+2\right)^2, 10=\frac{1}{n}\left(x_i-2\right)^2$
Subtracting, $8=\frac{1}{n}\left(\sum\left(x_i+2\right)^2-\sum\left(x_i-2\right)^2\right)$
$=\frac{8}{n}\sum x_i \Rightarrow \bar{x}=1$
Also $18=\frac{1}{n}\sum\left(x_i^2+4+4x_i\right)$
$=\frac{1}{n}\sum x_i^2+4+4\frac{1}{n}\sum x_i$
$=\frac{1}{n}\sum x_i^2+8$
So $10=\frac{1}{n}\sum x_i^2$
$\sigma_X^2=\frac{1}{n}\sum x_i^2-\left(\frac{1}{n}\sum x_i\right)^2=10-1=9$
Thus $\sigma_X=3$

Q.17 Answer:

11,13

Solution:

Mean of the given series is 5. If we have to transfer the given series into one having 12 as mean, we must take $d=x+7$**. The transformed terms are** $8, 9, 11, 13$**, 15,16. The two middle terms are 11,13.**

Q.18 Answer:

$3^{(n+1)/2}$

Solution:

G.M. $=\left(3.3^2\cdot 3^3\ldots 3^n\right)^{1/n}$
$=\left(3^{1+2+\ldots+n}\right)^{1/n}=\left(3^{\frac{n(n+1)}{2}}\right)^{1/n}$
$=3^{(n+1)/2}$

Q.19 Answer:

82,23

Solution:

Let the missing frequencies be f_1 **and** f_2**, respectively.**
\begin{array}{|ccc|} \ \hline \text { Class } & \text { Frequency }(f) & \text { Cum. Freq. (c.f.) } \\ \ \hline 10-20 & 180 & 180 \\ \ 20-30 & f_{1} & 180+f_{1} \\ \ 30-40 & 34 & 214+f_{1} \\ \ \hline 40-50 & 180 & 394+f_{1} \\ \ \hline 50-60 & 136 & 530+f_{1} \\ \ 60-70 & f_{2} & 530+f_{1}+f_{2} \\ \ 70-80 & 50 & 580+f_{1}+f_{2} \\ \ \hline \text { Total } & 685 & \\ \ \hline \ \end{array}
We have $580+f_1+f_2=685 \quad \Rightarrow \quad f_1+f_2=105$
As median 42.6 **lies in the class** $40-50$**, median class is** $40-50$**. We have that** $l=40, f=$

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$180, C = 214 + f_1$ **and** $h = 10$

$\text{Now,} \quad \text{Median } = l + \frac{h}{f}\left(\frac{N}{2} - C\right)$

$\Rightarrow \quad 42.6 = 40 + \frac{10}{180}\left(\frac{685}{2} - 214 - f_1\right)$

$\Rightarrow \quad 2.6 \times 18 = 342.5 - 214 - f_1 \Rightarrow f_1 = 81.7$

As frequency of an item is always a whole number, we take

$f_1 = 82.$

$\therefore f_2 = 105 - 82 = 23$

Q.20 Answer:

M.D. < S.D.

Solution:

M.D. $= \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$ **and S.D.** $= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

Let $a_i = |x_i - \bar{x}|$ **so (M.D.)** $^2 = \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)^2$ **and (S.D.** $)^2 = \frac{1}{n}\sum_{i=1}^{n} a_i^2$**. By Cauchy Schwarz inequality**

$\left(\sum_{i=1}^{n} a_i\right)^2 < n\sum_{i=1}^{n} a_i^2$

$\Rightarrow \quad \left(\frac{1}{n}\sum a_i\right)^2 < \frac{1}{n}\sum a_i^2 \Rightarrow$ **M.D.** $<$ **S.D.**

Q.21 Answer:

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Solution:

We know that if $d_i = \frac{x_i - A}{h}$ **then** $\sigma_x = |h|\sigma_d$**.**

In this case $-2x_i - 3 = \frac{x_i + 3/2}{-1/2}$ **so** $h = -1/2$**. Thus** $\sigma_d = \frac{1}{|h|}\sigma_x = 2 \times 3.5 = 7$

Q.22 Answer:

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Solution:

We know median of a arranged distribution is the term which middle most. So if observation greater than median is increased then it won't affect the relative position of the middle observation. Hence median will remain same. We know median of a arranged distribution is the term which middle most. So if observation greater than median is increased then it won't affect the relative position of the middle observation. Hence median will remain same.

Q.23 Answer:

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Solution:

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Let $\bar{x}$ be the mean wages of workers in the night shift, then

$60 = \frac{70 \times 54 + 30\bar{x}}{70+30}$

$\Rightarrow 600 = 7 \times 54 + 3\bar{x}$

$\Rightarrow \bar{x} = 200 - 7 \times 18 = 74$

Q.24 Answer:

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Solution:

Given there 2 a class of 20 students here we have to take spceial coses to find maximum marienee.

Take 10 students get 10 marks

10 students get 0 marks

$$\text{Then} \quad \text{mean} = \frac{10 + 10 + 10 + \cdots\cdots + 10 \text{ fimes}}{20}$$

$$= 5$$

$$\therefore \text{ variance} = \sum_{0}^{20} \frac{(5)^2}{20}$$

$$= \frac{20 \times 5^2}{20}$$

$$= \frac{20 \times 25}{20}$$

$$= 25$$

Q.25 Answer:

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Solution:

Mean of $(x_1+1)^2, \ldots, (x_{10}+1)^2$

$= \frac{1}{10}\sum_{j=1}^{10}(x_j+1)^2 = \frac{1}{10}\sum_{j=1}^{10} x_j^2 + \frac{2}{10}\sum_{j=1}^{10} x_j + 1$

$= \sigma^2 + \bar{x}^2 + 2 \times 2 + 1$

$= 9 + 4 + 4 + 1$

$= 18$

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